

Lecture 1

Software Types Lecture Notes

How to write an answer

Start with a definition. Explain the main features or functions in separate points. Add examples and a simple labelled diagram when appropriate. For a longer answer, include uses, advantages and limitations. In a comparison answer, compare both categories using the same points.

1 Software and its classification

Meaning of software

Software is a collection of programs and related instructions that tells a computer how to perform tasks. A program contains instructions that the computer processes. Software controls hardware operations and helps users complete activities such as writing documents, calculating marks and maintaining records.

Hardware and software

Hardware consists of the physical parts of a computer that can be touched, such as the keyboard, monitor, processor and printer. Software consists of the instructions and programs used by those parts. A printer is hardware; a word processor used to prepare the page is software. A saved document is data created using software.

Basis	Hardware	Software
Meaning	Physical computer components	Programs and related instructions
Examples	Keyboard, CPU, printer, SSD	Operating system, word processor, DBMS
Role	Performs physical operations	Directs operations and provides functions
Relationship	Needs instructions for useful work	Needs suitable hardware to execute

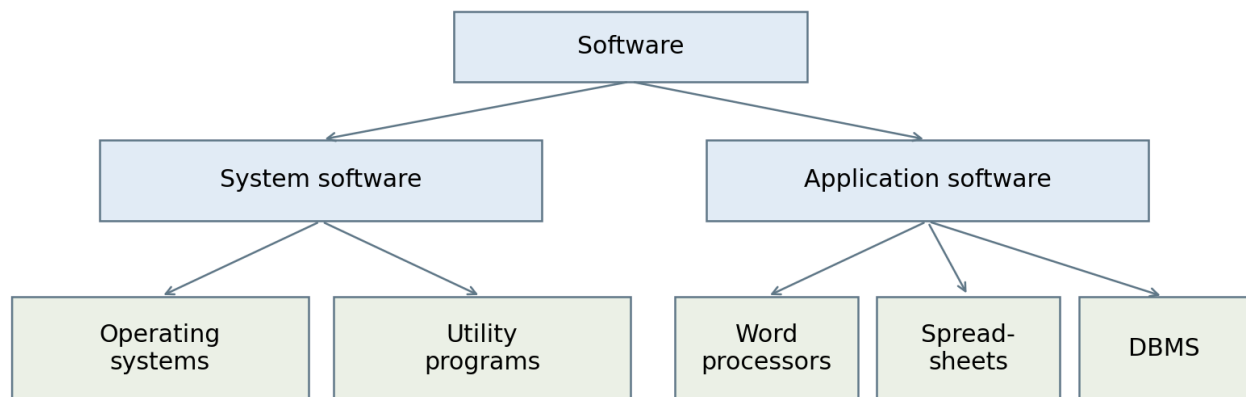


Figure 1 Classification of software

Two main categories

- **System software** manages the computer and provides a platform for applications. Operating systems and utility programs belong to this category in the syllabus.
- **Application software** helps users perform specific tasks. Word processors, spreadsheets and DBMS are the application categories studied here.

Simple example

When a student writes an assignment, the word processor helps create and format the document. The operating system manages the application's memory, file storage and interaction with devices. A backup utility can preserve a copy of the assignment. Each tool contributes a different function.

2 System software

Definition

System software is software that manages computer resources, supports the operation of the computer and provides services for application software. Resources include processor time, memory, storage and input/output devices. System software forms the supporting environment in which applications work.

Main characteristics

- Resource management: coordinates the use of hardware and storage.
- Support role: provides services that applications need to operate.
- Close connection with the system: works with devices and operating conditions.
- Essential platform functions: an operating system supports routine computer use.
- Background and visible work: some tasks run in the background, while others have user interfaces.

Important examples

- Operating system: manages resources and allows applications to run. Examples include Windows, macOS, Linux-based operating systems and Android.
- Utility program: performs a maintenance or protection task, such as backup, compression or malware scanning.
- Device driver: enables the operating system to work with particular hardware, such as a printer. Drivers provide useful supporting context beyond the named syllabus examples.

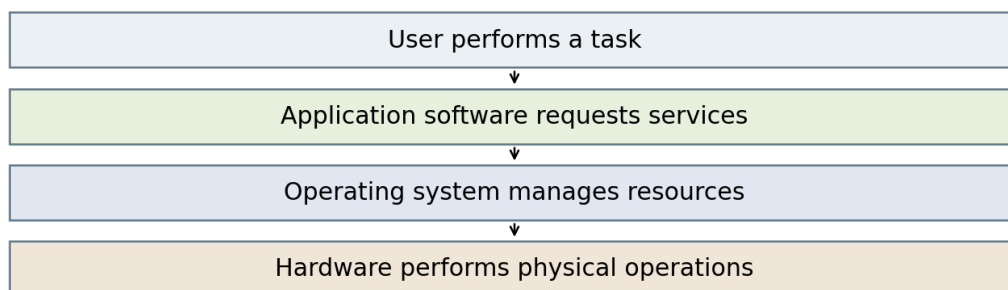


Figure 2 Simplified relationship between users applications the operating system and hardware

Why system software matters

Applications need a way to request memory, save files, display output and communicate with devices. The operating system provides these services and manages competing demands. Utilities support maintenance and protection. Without suitable system support, ordinary applications cannot use the computer effectively.

3 Operating systems

Definition

An operating system, or OS, is system software that manages hardware and software resources and provides services for application programs and users. It coordinates program execution, memory, files and devices. It also provides ways for a user to interact with the computer.

The OS as a manager

Suppose a student opens a word processor and a spreadsheet at the same time. Both need memory, processor time and access to storage. The operating system allocates resources and coordinates their use so that the programs can work together. It also receives keyboard input and helps deliver output to the screen or printer.

Main functions of an operating system

- Process management: starts, schedules and ends running programs. A running instance of a program is called a process.
- Memory management: allocates RAM to processes and tracks its use. It helps prevent one process from improperly accessing another process's memory.
- File management: organises files and folders, handles access and supports operations such as creating, opening, renaming and deleting files.
- Device management: coordinates input/output devices with the help of drivers. It handles requests to devices such as printers and storage drives.
- User interface: provides an interaction method, such as graphical windows or a command interface.
- Security and access control: manages accounts, permissions and other controls to help protect resources.
- Error handling and monitoring: records or reports problems and provides information about resource use.

Program and process

A program is a stored set of instructions. A process is a program in execution. A word processor installed on a drive is a program. When the user opens it and it runs, the operating system manages a process associated with it. A single application can also create several processes.

Multitasking

Multitasking means supporting more than one task during the same period. The OS schedules processor time among tasks; multiple cores can also perform work in parallel. For example, music can play while a document is being edited. Multitasking requires coordination rather than unlimited resources.

An operating system is system software that manages a computer's resources and provides an environment for applications. Its functions include process, memory, file and device management, user interaction and access control.

3 Operating system functions through examples

Function	Example	What the OS does
Process management	Word processor and browser are open	Schedules execution and tracks processes
Memory management	A spreadsheet opens a large file	Allocates memory and manages its availability
File management	A student saves Assignment.docx	Handles file storage and access through the file system
Device management	A document is printed	Coordinates the print request and device communication
User interface	A student clicks a folder icon	Receives the action and provides the requested view
Access control	Another user opens a private folder	Checks permissions according to configured rules

Saving a document step by step

1. The user selects Save in the word processor.
2. The application requests file services from the operating system.
3. The OS uses the file system and storage services to write the document data.
4. The storage device retains the data, and the application reports the save result.

This is a simplified description. The important point is that the application supplies the document task, while the operating system supplies underlying services. Saving a file to storage is different from keeping unsaved changes only in working memory.

Benefits of an operating system

- Makes hardware services available through a consistent environment.
- Allows users to run and switch between applications.
- Organises files and manages storage access.
- Coordinates devices and resources.
- Provides account and permission controls.

Limitations and responsibilities

- Uses some processor time, memory and storage itself.
- Requires compatible hardware, drivers and applications.
- Can contain defects or vulnerabilities and may require maintenance.
- Cannot guarantee that every application is correct or every file is safe.
- Resource management cannot overcome insufficient RAM, storage or processing capacity.

4 Operating system examples and interfaces

Examples of operating systems

Example	Typical environment	Easy explanation
Microsoft Windows	Desktop and laptop computers	Provides an environment for general computer applications
macOS	Apple Mac computers	Manages Mac hardware and supports Mac applications
Linux-based operating systems	Desktops, servers and embedded devices	Examples include Ubuntu; Linux is the underlying kernel
Android	Phones and other mobile devices	Supports mobile applications and device services
iOS	Apple iPhone	Provides the operating environment for iPhone applications

A kernel is the central part of an operating system that handles core operations. In everyday introductory usage, people often say “Linux operating system.” More precisely, distributions such as Ubuntu combine the Linux kernel with other components to form a complete operating environment.

Graphical user interface

A graphical user interface, or GUI, lets a user interact through visual elements such as windows, icons, menus and buttons. For example, double-clicking a document icon can open the file in its associated application. A GUI is often convenient for beginners because many available actions are visible.

Command line interface

A command line interface, or CLI, lets a user type commands to request actions. A terminal or command shell provides this form of interaction. Commands must follow the required syntax. A CLI can be useful for repeated tasks and administration, but beginners must learn the command names and options.

Point	GUI	CLI
Interaction	Click, tap or select visual controls	Type commands
Learning	Often easier to explore visually	Requires command knowledge
Typical use	Routine desktop tasks	Administration and automation
Example	Opening a folder through an icon	Entering a command to list files

5 Utility programs

Definition

A utility program is software designed to perform a specific maintenance, management or protection task for a computer or its data. Utilities support reliable operation by helping users protect files, inspect resources, manage storage or prepare data for transfer. They may be included with the OS or installed separately.

Utilities and the operating system

The operating system provides broad resource-management services. A utility focuses on a narrower task. For example, the OS manages files and storage access, while a backup utility makes copies that can help recover lost files. A utility can use OS services while carrying out its specialised work.

Major types

- **Antivirus or anti-malware utility:** scans for malicious software and can block, isolate or remove detected threats. It cannot guarantee detection of every threat.
- **Backup utility:** creates copies of selected data for later restoration.
- **Compression utility:** packages files and may reduce their size for storage or transfer. ZIP tools and 7-Zip are examples.
- **Disk cleanup utility:** identifies temporary or unnecessary data that may be removed to free storage.
- **Disk checking utility:** checks aspects of a file system or storage health and may repair supported errors.
- **Storage optimisation utility:** performs operations suited to the storage technology. Traditional HDD defragmentation and SSD optimisation are different operations.
- **System monitoring utility:** displays information about processes, processor use, memory or storage.

Practical examples

Situation	Suitable utility	Purpose
A teacher wants a recoverable copy of notes	Backup utility	Preserve data for restoration
Several files must be sent together	Compression utility	Create an archive
Storage is almost full	Disk cleanup utility	Identify removable temporary data
A program is using much memory	System monitoring utility	Inspect resource usage
A file may contain malware	Anti-malware utility	Scan and handle detected threats

Classification point

A utility helps manage or maintain a system even when it has a visible window. Software classification depends mainly on purpose, not on whether the user can click buttons in it.

5 Utility programs explained in detail

Backup and restore

Backup means creating a copy of data so that it can be recovered if the original is lost or damaged. Restore means recovering data from that copy. For example, a teacher may back up examination notes to another device or an appropriate remote service. Merely moving the only copy is not a backup.

- A useful backup should preserve the required data in a recoverable form.
- A separate copy reduces dependence on the original device.
- Restoration should be checked so that the backup is known to be usable.
- Automatic synchronisation can help access files, but it may also copy unwanted changes or deletions; it is not always equivalent to a protected backup.

Compression

Compression represents data in a form that may occupy less space. File archives can group several files together. General-purpose archive compression is normally lossless: the original file contents can be reconstructed. Already compressed photos or videos may shrink very little. Compression does not automatically encrypt or protect a file from unauthorised access.

Disk cleanup and disk checking

Disk cleanup frees space by removing selected unnecessary data. Disk checking examines file-system or storage conditions and can repair some supported problems. They have different purposes: cleanup deals mainly with removable content, while checking deals with consistency or faults. Neither guarantees recovery of all lost data.

HDD defragmentation and SSD optimisation

On a hard disk drive, parts of a file can occupy separate physical locations. Defragmentation rearranges data to reduce fragmentation and may improve some access patterns. Solid-state drives use different technology and do not benefit from traditional mechanical-seek defragmentation in the same way. Storage maintenance should be appropriate to the drive type.

Advantages and limitations

- Advantages: supports maintenance, protects recoverable copies, assists troubleshooting and helps manage storage.
- Limitations: cannot fix every hardware fault, cannot guarantee complete security and may require correct configuration.
- Care is needed: deleting the wrong files or selecting an unsuitable repair operation can cause problems.

6 Application software

Definition

Application software is software designed to help users perform specific tasks. It uses the services of system software to carry out activities such as writing a report, calculating an expense total or retrieving student records. Word processors, spreadsheets and DBMS are application categories in this syllabus.

Characteristics

- Task focus: each application supplies functions for a particular activity or group of activities.
- User interaction: applications commonly accept input and present results through an interface.
- Dependence on the environment: applications need appropriate OS or runtime support.
- Data creation and processing: applications produce documents, worksheets or database records.
- Different forms: applications may be installed locally or accessed through a browser.

Examples

Application type	Purpose	Examples
Word processor	Create and format text documents	Microsoft Word, Google Docs, LibreOffice Writer
Spreadsheet	Organise tabular data and perform calculations	Microsoft Excel, Google Sheets, LibreOffice Calc
DBMS	Create, store, retrieve and manage database data	Microsoft Access, MySQL, PostgreSQL, SQLite

General purpose and customised applications

General-purpose applications support common tasks for many users, such as document writing and numerical work. Customised applications are developed or adapted for particular requirements, such as a college admission system. A customised application can use a DBMS to maintain records and a word processor to prepare related documents.

Advantages

- Improves productivity by providing tools for a defined task.
- Helps present and process information consistently.
- Reduces repeated manual work through reusable features.
- Allows users to save, revise and share their work.

Limitations

- Requires users to learn relevant features and workflows.
- May require suitable hardware, licences or network access.
- File formats and application versions can affect compatibility.
- Incorrect input or settings can produce incorrect output.

A browser-based application still relies on system software. The browser runs within an operating environment, and the online service also uses computers and software elsewhere. Accessing an application through the internet does not remove its need for underlying system support.

7 Word processors

Definition

A word processor is application software used to create, edit, format, save and print text-based documents. Examples include Microsoft Word, Google Docs and LibreOffice Writer. It is suitable for letters, applications, reports, assignments and notices.



Figure 3 Typical stages of preparing a document with a word processor

Main features

- Text entry and editing: type, insert, delete, copy and move text.
- Character formatting: change font, size, bold, italic and text colour.
- Paragraph formatting: adjust alignment, indentation, line spacing and lists.
- Page layout: set margins, page size and orientation.
- Tables and objects: insert tables, pictures and supported shapes.
- Proofing: spelling and grammar tools help identify possible mistakes.
- Find and replace: locate text and replace repeated words or phrases.
- Headers and footers: add repeated information and page numbers.
- Mail merge: combine a template with a data source to create personalised documents.
- Saving and output: store the document, export supported formats and print.

Editing and formatting

Editing changes the content. Replacing an incorrect student name is editing. Formatting changes the appearance. Making the heading bold or centring it is formatting. Both operations are important, but they solve different problems.

Common formats

DOCX and ODT are common editable document formats. PDF is often used to distribute a document with a stable page layout. Plain text stores basic characters but does not preserve rich layout. Available formats depend on the application. Changing the filename extension alone does not properly convert a file.

Example

A student prepares a leave application by typing the recipient, subject and request. The student corrects spelling, sets appropriate paragraph spacing, saves the document and prints it. The word processor provides the writing task; the operating system provides storage and printing services.

7 Word processor uses advantages and limitations

Uses

- Writing formal letters and leave applications.
- Preparing student assignments and project reports.
- Creating notices, invitations and certificates.
- Preparing resumes and meeting minutes.
- Producing personalised letters through mail merge.

Advantages

- Easy correction: errors can be changed without rewriting the whole document.
- Professional layout: formatting helps make documents readable.
- Reusable content: templates and copying save time.
- Efficient proofing: tools help locate spelling and some grammar problems.
- Convenient storage: documents can be saved and revised later.
- Multiple output options: documents can be printed or exported in supported formats.

Limitations

- Proofing is imperfect: software may miss mistakes or suggest unsuitable changes.
- Formatting requires judgement: too many fonts or inconsistent spacing can reduce readability.
- Compatibility can affect layout: another application may display some features differently.
- Unsaved changes can be lost: saving and recoverable copies remain important.
- Complex structured records are better handled by appropriate data tools.

Mail merge explained

Mail merge combines a main document with a list of changing values. For example, one certificate template contains fields for Name and Roll Number. A data source supplies those values for each student. The application produces a personalised version for each record. This avoids manually retyping the same text, but the source data must be accurate.

8 Spreadsheet applications

Definition

A spreadsheet is an electronic grid of rows and columns used to organise and process data. Spreadsheet application software creates this grid and provides calculations, formatting and data-management tools. Examples are Microsoft Excel, Google Sheets and LibreOffice Calc.

Basic terms

- Row: a horizontal line of cells, usually identified by a number.
- Column: a vertical line of cells, usually identified by letters.
- Cell: the intersection of a row and a column.
- Cell address: a reference such as B2, meaning column B and row 2.
- Worksheet: one grid or sheet in a spreadsheet document.
- Workbook: a spreadsheet file that can contain several worksheets.
- Formula: an expression used to calculate a result, commonly beginning with =.

Example marks sheet

Row	Column A Name	Column B Theory	Column C Practical	Column D Total
1	Name	Theory	Practical	Total
2	Riya	70	20	=B2+C2 gives 90
3	Aman	60	18	=B3+C3 gives 78

In this example, B2 contains 70 and C2 contains 20. The formula =B2+C2 in D2 produces 90. If a referenced value changes, the result normally updates when automatic calculation is enabled. The formula refers to cells instead of requiring the total to be typed again.

Main features

- Data entry: store text, numbers, dates and formulas in cells.
- Calculation: formulas and functions calculate totals, averages and other results.
- Formatting: apply fonts, borders, alignment and number display formats.
- Sorting: rearrange complete records in a chosen order.
- Filtering: temporarily display records that satisfy conditions.
- Charts: represent numerical patterns visually.
- Multiple worksheets: organise related information within a file.

Uses

Spreadsheets are used for marks sheets, attendance summaries, household budgets, stock lists and sales calculations. They are especially useful when data need frequent calculations or small, easily inspected tabular analyses.

8 Spreadsheet advantages limitations and examples

Advantages

- Organised data: rows and columns make related information easy to inspect.
- Fast calculation: formulas reduce repeated arithmetic.
- Automatic recalculation: results can update when input values change.
- Easy editing: entries can be corrected or extended.
- Sorting and filtering: records can be arranged or narrowed for inspection.
- Visual presentation: charts and formatting help explain numerical information.

Limitations

- Incorrect formulas or data can produce incorrect results.
- Large or complicated workbooks can become difficult to audit and maintain.
- Accidental overwriting can damage formulas or records.
- Sorting only one column can break relationships between names and values.
- Complex multiuser records and strict relational rules may require a DBMS.
- Security and recovery depend on appropriate access controls and saving practices.

Sorting and filtering

Sorting changes the order of records. Sorting marks from highest to lowest places the highest-scoring student first. Filtering displays only matching records while temporarily hiding others. Filtering marks to show values at least 80 focuses on high scores. Neither operation should delete records merely as part of its normal purpose.

Formula and function

A formula is a calculation expression, such as `=B2+C2`. A function is a named, predefined operation used within a formula, such as `SUM` or `AVERAGE`. For example, `=SUM(B2:B5)` adds values in the selected range. The spelling, separators and available functions can depend on the application and locale.

Worked budget example

Item	Amount
Books	600
Travel	400
Stationery	200
Total	1200

A spreadsheet can store the three amounts and calculate the total with a formula. If Travel changes to 500, the total becomes 1300 after recalculation. A manually typed total would need a separate correction.

9 Database management systems

Meaning of a database

A database is an organised collection of related data. For example, a college database may store students, courses and enrolments. A relational database represents data in tables and uses relationships between them. Other kinds of databases also exist; the table examples here introduce the relational approach.

Definition of a DBMS

A database management system, or DBMS, is software that allows users and applications to define, store, retrieve, update and manage database data. It controls how data are accessed and supports rules for managing them. Examples include Microsoft Access, MySQL, PostgreSQL and SQLite.



Figure 4 A DBMS processes requests and manages access to stored data

Database and DBMS are different

The database is the organised data, while the DBMS is the software that manages those data. A table of student records belongs to the database. PostgreSQL is a DBMS that can manage such tables. SQL is a language used with many relational DBMS products; SQL itself is not the stored database or the DBMS product.

Basic terms

Term	Meaning	Student example
Table	Collection of related records	Students table
Record or row	One complete entry	Details of one student
Field or column	One attribute	Name or Roll Number
Primary key	Field or combination that uniquely identifies a record	A unique Student ID
Query	Request to retrieve or process data	Find students enrolled in a course

Example student table

Student ID	Name	Town
101	Riya	Bhoranj
102	Aman	Hamirpur
103	Tanvi	Nadaun

Each row represents one student, while each column represents one field. Student ID can be used as the primary key if it is unique and does not contain missing values. A name alone is usually unsuitable because different students may have the same name.

9 DBMS functions relationships and applications

Main functions

- Data definition: creates tables and specifies fields, data types and rules.
- Data storage: organises records for retrieval and modification.
- Data retrieval: processes queries to locate required information.
- Data modification: adds, changes and removes records through controlled operations.
- Integrity controls: applies rules that help keep data valid and consistent.
- Access control: restricts permitted operations according to the system's capabilities and configuration.
- Concurrent use: suitable DBMS products coordinate changes made by multiple users.
- Backup and recovery support: provides or supports methods to restore data after specified failures.

Relationships between tables

A college may store student details in a Students table and course details in a Courses table. An Enrolments table links students with their courses. Instead of repeatedly typing a student's full address beside every course, records can refer to the Student ID. Properly designed relationships reduce unnecessary duplication and help maintain consistency.

Students	Courses	Enrolments
Student ID	Course ID	Student ID
Name	Course Name	Course ID
Town	Credits	Academic Year

This table shows a simplified structure, not actual records. A foreign key is a field or set of fields that refers to a key in a related table. In a suitably defined design, Enrolments.Student ID refers to Students.Student ID. Such rules can help stop enrolments from referring to nonexistent students.

Data integrity

Data integrity means maintaining correct, valid and consistent data. Examples include requiring unique Student IDs, rejecting a missing compulsory value or allowing only valid course references. A DBMS can enforce defined rules, but it cannot know that every real-world fact entered by a user is true.

Transactions

A transaction is a group of operations treated as one logical unit. For example, a fee payment update and its receipt record should remain consistent. Suitable transactional facilities help ensure that related changes complete together or are handled as a failure rather than leaving an incomplete state.

Applications

- College admissions and student records.
- Banking and payment records.
- Library catalogues and book issue records.
- Hospital appointments and patient records.
- Business stock, customers and sales information.

9 DBMS advantages limitations and model answer

Advantages

- Organised storage: related records are managed systematically.
- Reduced unnecessary duplication: good design avoids repeating the same details in many places.
- Improved consistency: defined rules help keep related data in agreement.
- Efficient retrieval: queries locate required records without manually reading every entry.
- Controlled sharing: suitable DBMS products support multiple users with access restrictions.
- Recovery support: backup and transaction facilities can help manage failures.

Limitations

- Complexity: database design and administration require knowledge.
- Setup and maintenance: some systems need substantial configuration and ongoing work.
- Resource requirements: large databases may need significant storage and processing capacity.
- Possible cost: hardware, support or commercial licences may add expense, although free products also exist.
- Dependence on the database service: disruption can affect many applications and users.
- Security is not automatic: weak access settings or application flaws can still expose data.

10 System and application software compared

Basis	System software	Application software
Main purpose	Manages and supports the computer	Helps perform a user task
Main focus	Resources and underlying services	Documents, calculations or business records
Relationship	Provides the operating environment	Uses the operating environment
Examples	Operating system, backup utility, driver	Word processor, spreadsheet, DBMS
User interaction	May run in the background or provide a visible interface	Commonly provides task-specific interaction
Need	An OS is normally required for routine general-purpose operation	Particular applications depend on the user's needs

OS and utility compared

Basis	Operating system	Utility program
Scope	Broad management of the operating environment	A specialised support task
Function	Processes, memory, files and devices	Backup, scanning, cleanup or inspection
Example	Windows or Ubuntu	A backup tool or compression utility
Packaging	Core environment for routine use	Can be built in or installed separately

10 Choosing between a word processor spreadsheet and DBMS

Basis	Word processor	Spreadsheet	DBMS
Main purpose	Prepare text documents	Process tabular and numerical data	Manage structured database records
Typical structure	Pages and paragraphs	Cells, rows and columns	Tables and defined relationships
Main tools	Text editing and page formatting	Formulas, functions and charts	Queries, rules and data operations
Best example task	Write a college notice	Calculate class marks	Maintain student enrolments
Examples	Word, Docs, Writer	Excel, Sheets, Calc	Access, MySQL, PostgreSQL

Model examination answers

Question 1 Explain the types of software

Software is a collection of programs and related instructions that directs computer operations. According to purpose, it can be classified into system software and application software.

- System software manages computer resources and provides services for applications. An operating system manages processes, memory, files and devices. Utility programs perform specialised tasks such as backup, compression and maintenance.
- Application software helps users perform tasks. A word processor prepares documents, a spreadsheet performs tabular calculations, and a DBMS manages organised records.

For example, Windows provides the operating environment, a backup utility preserves copies, Word prepares a report, Excel calculates marks and a DBMS maintains student records. System and application software work together to make the computer useful. Draw Figure 1 and explain each branch to extend the answer.

Question 2 Explain the functions of an operating system

An operating system is system software that manages resources and provides services to programs and users. Its important functions are:

- Process management: starts and schedules programs in execution.
- Memory management: allocates and tracks RAM used by processes.
- File management: organises stored files and controls access.
- Device management: coordinates input/output devices using drivers.
- User interface: allows interaction through graphical controls or commands.
- Access control: applies accounts and permissions to supported resources.

For example, while a student edits a document and plays music, the OS coordinates both activities and manages memory, output and storage. It provides the services that both applications require.

Question 3 Explain utility software with examples

Utility software performs specialised tasks that maintain, protect or manage a computer and its data. A backup utility copies data for recovery, a compression utility creates archives, an anti-malware utility handles detected threats, and a cleanup utility removes selected unnecessary data. Utilities may be included in an OS or installed separately. They assist reliable operation but cannot solve every hardware fault or guarantee complete protection.

More model answers and answer writing guidance

Question 4 Explain application software and its examples

Application software helps users complete specific tasks using the services of the operating environment. The important syllabus categories are word processors, spreadsheets and DBMS.

- Word processors create, edit, format, save and print documents. Examples include Word, Google Docs and Writer.
- Spreadsheets organise data in cells and support formulas, functions, sorting, filtering and charts. Examples include Excel, Google Sheets and Calc.
- DBMS products define and manage organised records through storage, queries, updates and rules. Examples include Access, MySQL and PostgreSQL.

These applications improve productivity but require accurate input and suitable use. A college can use a word processor for notices, a spreadsheet for marks calculation and a DBMS for linked student and course records.

Question 5 Differentiate a database and a DBMS

A database is an organised collection of related data. A DBMS is software that manages those data. For example, a collection of student and course tables forms part of a college database, while PostgreSQL can be the DBMS used to store and query them. The data and the managing program have different roles. SQL is a language used to communicate operations to many relational DBMS products.

Question 6 Explain why system and application software need each other

System software supplies the environment and resources for applications. Application software uses those services to complete user tasks. When a document is saved, the word processor manages its contents and requests storage services. The operating system manages file access and hardware communication. The layers therefore perform different but connected functions.

Adapting an answer to the question

- Definition question: give the meaning and one relevant example.
- Functions question: state each function and explain its operation.
- Advantages question: give benefits with clear reasons.
- Comparison question: use the same bases for both categories.
- Long answer: combine definition, main points, examples and a labelled diagram.

Write in your own clear words. Avoid using an example as the whole definition: “Excel is a spreadsheet” names an example, while “a spreadsheet organises data in cells and supports calculations” explains the concept.

Short answer revision and practice questions

What is software

Programs and related instructions that direct computer operations.

What is system software

Software that manages resources and supports the operating environment.

What is application software

Software that helps a user perform a specific task.

What is an operating system

System software that manages resources and provides services to programs and users.

What is a utility program

A program for a specialised maintenance, management or protection task.

What is a word processor

An application for creating, editing, formatting and outputting documents.

What is a spreadsheet

An electronic grid used to organise data and support calculations.

What is a DBMS

Software that defines, stores, retrieves, modifies and manages database data.

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